SRS/SDD Self-Review Report – Review 2

1. Run Information

Project code	FA26SE032
Project name	Sales Collaborator and Affiliate Network Management System (SCANMS)
Team	Nguyễn Thành Thắng (Leader - SE184251), Nguyễn Đình Tuấn (SE182540), Nguyễn Phú Quý (SE180104), Phan Xuân Thịnh (SE184527)
Supervisor	ThS. Tôn Thất Hoàng Minh
AI tool and model	Antigravity IDE – Claude Opus 4.6 (Full-Repository Coding Agent)
Run date	06/11/2026
Repository tag / commit	review-2 (commit 99efe6a)
Registration form version used	Version 2.0 (Review 2 Proposed Final Scope)
Version	2.0
Date	06/11/2026

2. Standard Prompt

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You are reviewing the SRS and SDD of a software capstone project against its
registration form and source code. Report only; do not change any file.

Inputs: the registration form, SRS, SDD (with diagrams), source code and
database script in this repository.

Check:
1. Scope: FRs in the SRS not covered by the registration form, and
   registration items with no FR.
2. Requirements: FRs without an actor or testable acceptance criteria;
   NFRs without metric, threshold and measurement method.
3. Data: entities missing from the Logical ERD; ERD entities or attributes
   missing from the database script, and tables not in the ERD.
4. States: for each main entity, states used in code but missing from its
   state machine; transitions with no triggering FR; states with no exit.
5. Flows: main business flows in activity diagrams with no implementation
   or no test.
6. Architecture: components or third-party services in code but not in the
   architecture diagram, and the reverse.
7. Design vs code: modules, APIs or business rules implemented differently
   from the SDD.

Output one table: ID | Category | Location | Finding | Severity
(High/Medium/Low) | Evidence. Then list the 5 most important findings.
If something cannot be verified from the repository, say so; do not guess.
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3. Findings and Team Response

ID	Category	Location	Finding	Severity	Evidence
SR-01	Data	docs/database/DATABASE_DESIGN.md & docs/architecture/SAD_DOCUMENT.md vs backend/prisma/schema.prisma	Logical ERD in early SDD depicts 21 tables (and lists legacy role SHOP_STAFF), whereas schema.prisma implements 69 models and 41 enums including ProductPricePromotion, ExclusiveDealProposal, AttributionSession, ReturnRequest, ReturnShipment, ReturnRefund, ReturnDispute, MonthlyBonusResult, FinancePosting, and LiveSessionProduct under the 5 UserRole values.	**High**	docs/database/DATABASE_DESIGN.md:10-60; backend/prisma/schema.prisma:13-19, 319-21
SR-02	Architecture	docs/architecture/SAD_DOCUMENT.md vs frontend/, Mobile/, backend/	Architecture diagram labels client portals as Next.js and async worker as BullMQ, whereas the repository implements React 19 + Vite (frontend/), React Native 0.88 + Expo SDK 58 (Mobile/) via GET /api/mobile/bootstrap, @nestjs/schedule cron workers, PayOS, BIDV Sandbox, GHN Sandbox, and Cloudinary.	**High**	docs/architecture/SAD_DOCUMENT.md:14-41; frontend/package.json; Mobile/package.json; backend/src/modules/mobile/mobile.controlle
SR-03	Scope	Proposed Registration v2.0 vs backend/prisma/schema.prisma & Mobile/src/features/workspace/policy.ts	Standalone promotional portals were eliminated from Registration v2.0 and redirected out of Web routes (AppRoutes.tsx:203,231,266), but legacy Campaign model/module and a Mobile workspace menu entry remain dormant in the repository (with rate parameter fixed at 0 in active order settlement).	**Medium**	frontend/src/routes/AppRoutes.tsx:203-266; backend/prisma/schema.prisma; Mobile/src/features/workspace/policy.ts:68-71
SR-04	Flows	Registration v2.0 Deliverables vs Mobile/docs/ROLE_WORKFLOWS_2026-11-03.md	Both Web Portal and Expo Android Mobile App are in scope, and 5-role Mobile operations (catalog, KYC, chat, VIP deals, samples, pricing, moderation, GHN orders, returns, payouts) are implemented, but native Mobile PayOS checkout and Live room playback are not yet ported on Android.	**High**	Mobile/docs/ROLE_WORKFLOWS_2026-11-03.md:22,88; Mobile/src/features/cart/CartScreen.tsx
SR-05	Architecture	Registration v2.0 Payouts vs backend/src/modules/payouts/bidv-payout.adapter.ts	Collaborator bank disbursement uses BidvPayoutAdapter (PAYOUT_PROVIDER=BIDV_SANDBOX with idempotency key and inquiry reconciliation) and VietQR XLSX batch export rather than licensed commercial bank host-to-host production rails.	**High**	backend/src/modules/payouts/bidv-payout.adapter.ts:15-96; docs/BIDV_SANDBOX
SR-06	Design vs code	Early SDD Commission Spec vs backend/src/modules/commissions/commission-policy.ts & commission.constants.ts	Code enforces Best-Benefit Precedence min(25%, max(VIP Deal, 10%+Tier, 10%) + Live), COMMISSION_HOLD_DAYS = 7 post-delivery hold, and conditionalDealEscrowBalance (DealBonusStatus) instead of early SDD's additive formula and 14-day hold.	**Medium**	backend/src/modules/commissions/commissio policy.ts:37-47; commission.constants.ts:3; wallets.service.ts:239-307
SR-07	States	Early SDD State Diagrams vs backend/prisma/schema.prisma	Code implements richer entity lifecycles than early 4-state SDD charts: ExclusiveDealStatus includes PENDING_ADMIN_APPROVAL (>16% gate); SampleRequestStatus has 11 states (including VIDEO_SUBMITTED, REVISION_REQUIRED, OVERDUE); ReturnRequestStatus has 19 states covering both EXCHANGE and REFUND.	**Medium**	backend/prisma/schema.prisma:67-79, 104-13 213-218; backend/src/modules/returns/return state.machine.ts

ID	Category	Location	Finding	Severity	Evidence
SR-08	Design vs code	Early Registration/DB Spec vs backend/src/modules/auth/auth.service.ts & Mobile/src/services/auth/	DATABASE_DESIGN.md:54 mentions bcrypt, whereas auth.service.ts hashes passwords with Argon2id, coerces public SHOP_MANAGER signup to CUSTOMER, assigns starting KOC tier 'Đồng' (Bronze 10%), and Mobile stores Bearer accessToken in Expo SecureStore without a refresh endpoint.	**Medium**	backend/src/modules/auth/auth.service.ts:182 Mobile/docs/MOBILE_ARCHITECTURE.md:55-6
SR-09	Design vs code	Early Payout Spec vs backend/src/modules/payouts/payouts.service.ts & payout-tax.service.ts	Code strictly restricts withdrawals to UserRole.COLLABORATOR (blocking SHOP_MANAGER with 403), enforces DEFAULT_MIN_PAYOUT_AMOUNT = 200,000 VND, and withholds a flat 10% PIT rate (PAYOUT_TAX_THRESHOLD = 0.00, PAYOUT_TAX_RATE = 0.10) posted to double-entry FinancePosting.	**Medium**	backend/src/modules/payouts/payouts.service 115; payout-tax.service.ts:4-22; withdrawal-policy.service.ts:5
SR-10	Requirements	SRS Rate Limiting & AI Spec vs backend/src/core/cache/cache.service.ts & modules/ai/	Redis rate limiting uses an in-memory LRU Map fallback on Redis disconnect, and AI Partner Matchmaking / Fraud Sentinel combine deterministic 4-axis heuristic scoring with optional LLM explanation fallback rather than training custom neural networks.	**Low**	backend/src/core/cache/cache.service.ts; backend/src/modules/ai/ai-recommendation.service.ts; ai-fraud.service.ts

4. Top 5 Findings (as reported by the AI)

#	Finding ID	Summary
1	**SR-01**	SDD Logical ERD Outdated vs Prisma — Early 21-table ERD must be regenerated to reflect the 69 normalized Prisma models and 5 exact UserRole values.
2	**SR-02**	Architecture Stack Drift in SAD — SAD diagrams referencing Next.js/BullMQ must be updated to React 19 + Vite, React Native/Expo SDK 58 (/api/mobile/bootstrap), @nestjs/schedule, PayOS, GHN, and BIDV Sandbox.
3	**SR-04**	Cross-Platform Mobile Checkout/Live Parity — 5-role Android app covers core affiliate & operations workflows while Customer PayOS checkout and Live playback remain Web-hosted.
4	**SR-05**	Sandbox Interbank Payout Boundary — Collaborator withdrawals execute real DB row locks and 10% PIT double-entry postings but disburse via BidvPayoutAdapter sandbox rather than licensed bank rails.
5	**SR-06**	Commission & Escrow Policy Alignment — Production code enforces non-additive Best-Benefit Precedence (cap 25%), 7-day post-delivery hold, and conditionalDealEscrowBalance for VIP KPIs.

5. Summary

Severity	Total	Fixed	Kept	Planned
High	4	0	1	3
Medium	5	4	0	1
Low	1	0	1	0
Total	10	4	2	4